AI

Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[18CS7AD01]**

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

ARTIFICIAL NEURAL NETWORK AND DEEP LEARNING

Time: 3 Hours **Maximum Marks: 70**

Instructions: i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1. a. Explain Neural Network architecture and discuss its advantages. **07**

b. Explain single layer perceptron, multilayer perceptron, perceptron learning rule with suitable example. **07**

2. a. What is Associate Memory Network? Describe two types of associate memory. **07**

b. Explain the Hebb Rule, Outstar learning rule, Delta Rule for Pattern Association **07**

3. a. **07**

3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

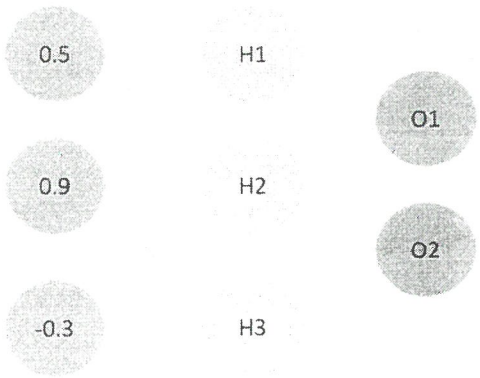
Consider this matrix as an image and find out the output of convolution layer where the size of Gabor filter is 3x3, which is mentioned below.

1	0	-1
1	0	-1
1	0	-1

b. Explain the working process of KNN along with advantage and disadvantage. **07**

OR

4. a. **07**



H1 Weights = (1.0, -2.0, 2.0)

H2 Weights = (2.0, 1.0, -4.0)

H3 Weights = (1.0, -1.0, 0.0)

O1 Weights = (-3.0, 1.0, -3.0)

O2 Weights = (0.0, 1.0, 2.0)

Draw an ANN and calculate the value of O1 and O2

b. What is back propagation algorithm? Explain it's working process. **07**

5. a. Explain Word Embedding and Word2Vec **07**

b. Explain Recurrent Neural Network (RNN). **07**

OR

6. a. What is GAN and explain its applications **07**

b. Explain the working process of auto encoder and give its application. **07**

7. a. Explain Genetic algorithm as optimizer. Describe all 5 phases of it. **07**

b. What is GRU? Explain with the architecture. **07**

OR

8. a. Explain the difference Between Genetic Algorithm and Traditional Algorithm along with suitable example **07**

b. Why do the gradients even vanish/explode? How could we find out that the model is suffering from Exploding/Vanishing gradient problem? **07**